Mohd Qasim

4th Year Undergraduate
Department of Chemical Engineering

Email : b23ch1027@iitj.ac.in
Phone : +91-6006993954

Academic Qualifications

Year	Degree/Certificate	Institute	CPI/%
2023 - Present	B.Tech	Indian Institute of Technology, Jodhpur	5.8/10
2022	CBSE(XII)	Jawahar Navodaya Vidyalaya (JNV), Budgam	72.6%
2020	JKBOSE(X)	Mutahhary Public High School Saliskote, Kargil	88.8%

Scholastic Achievements

- Secured **All India ST Rank 1095** in **JEE Advanced 2023** among the 1.2 Lakh shortlisted candidates.
- Secured **Rank** in "Kashmir Super 50" Entrance Exam conducted by CSRL.

Key Projects

- **Catalytic Synthesis of 4-Methoxypropiophenone via Friedel–Crafts Acylation**
  Mentor: Prof. Sumit Kamal, Department of Chemical Engineering.
  - Assisted in the synthesis of 4-methoxypropiophenone (MPP) through Friedel–Crafts acylation of anisole with propionic anhydride, employing Amberlyst-15 as a solid acid catalyst.
  - Performed time-based sampling of the reaction mixture to track product formation under varying catalyst loadings.
  - Operated a controlled-temperature 4-neck glass reactor system, ensuring uniform agitation and accurate thermal conditions.
  - Followed standard laboratory protocols for sample collection, catalyst handling, and reaction monitoring.
  - Gained hands-on experience with heterogeneous catalysis, aromatic substitution reactions, and process-oriented lab work.
- **Synthesis of Zeolite from Agricultural Waste for Water Purification**)
  Mentor: Prof. Sumit Kamal , Department of Chemical Engineering.
  - Synthesized zeolite using rice husk as a low-cost silica source through controlled hydrothermal process
  - Characterized structural and adsorption properties using advanced techniques (e.g., XRD, SEM, FTIR).
  - Applied synthesized zeolite for fluoride removal from drinking water and evaluated adsorption performance.
- **Named Entity Recognition (NER)**
  Mentor: Prof. Avinash Sharma , Department of Computer Science Engineering.
  - Built an end-to-end Named Entity Recognition (NER) system using Python and scikit-learn, focusing on entity extraction from unstructured text.
  - Conducted thorough data preprocessing, including handling noisy characters, missing values, and sentence segmentation from raw token-level data.
  - Engineered custom features (e.g., token patterns, casing, word length) and trained a Random Forest classifier for sequence tagging.
  - Evaluated model with NER-specific metrics (precision, recall, F1-score) using the nervaluate library; plotted evaluation results by entity type.
  - Utilized seaborn, matplotlib, and pandas to present results and trends, enhancing model explainability.

Technical Skills

- **Programming Languages:** C,C++,Python.
- **Software and Libraries:** Git, Github, OOPS, Flask, Aspen plus
- **Machine Learning Libraries:** Scikit-Learn, PyTorch, TensorFlow, Keras

Relevant Courses

Multivariable and vector calculus	Linear algebra and Differential equations
Introductory quantum chemistry and spectroscopy	Molecular Engineering
Introduction to computer science	Introduction to machine learning
Fluid Mechanics	Probability Statistics and Stochastic process
Thermodynamics	Material and Energy balance
Mass Transfer	Chemical Reaction Engineering